

# Care Access Navigator

Simulated Malware Incident Report | HealthLine Regional Medical Center (HRMC)

**Status:** SIMULATED / DRAFT | **Course Rubric:** Malware Analysis & MITRE ATT&CK Mapping | **Capstone**

**Week:** Week 2 (supplemental exercise)

## IN PLAIN TERMS

A workstation at HRMC was infected with malware that quietly records keystrokes and steals saved passwords, then sends that stolen data back to the attacker — most likely through email or a messaging-bot channel, which is how this malware family typically operates. All details below are simulated for coursework using a real, publicly catalogued malware sample as the technical reference point.

## 1. Incident Summary

| Field                          | Detail                                                           |
|--------------------------------|------------------------------------------------------------------|
| Incident ID                    | INC-W2-001 (simulated)                                           |
| Date/Time Detected (simulated) | Week 2, Day 4 — <fill from your test timeline>                   |
| Reporting Source               | Endpoint EDR alert + SIEM correlation (simulated)                |
| Malware Family                 | AgentTesla (MSIL-based keylogger / infostealer)                  |
| Sample Hash (SHA256)           | 1c72369e2543722e89be3a14448d0fdc1664f4638d91d67720dddb7ff73cb736 |
| VirusTotal Detection Ratio     | 34 / 70                                                          |
| Classification                 | Commodity infostealer / credential harvester                     |

## 2. Affected Asset (Simulated Mapping)

| Field            | Detail                                                                                                       |
|------------------|--------------------------------------------------------------------------------------------------------------|
| Asset ID         | AST-03 (adjust to match your asset inventory)                                                                |
| Asset Type       | Workforce endpoint (Windows workstation)                                                                     |
| Department       | Patient Registration / Billing (adjust to fit your AST-## entry)                                             |
| Data Sensitivity | ePHI-adjacent — endpoint has access to Billing App and shares a network segment with systems processing ePHI |

## 3. Malware Profile

**AgentTesla** is a long-running, widely distributed .NET-based infostealer/keylogger, commonly delivered via phishing attachments (macro-enabled Office documents, or malicious archive/executable attachments) — the same delivery path already simulated in the Part 2 SEToolkit exercise, giving this incident a coherent phishing-to-infection chain.

**Documented capabilities (publicly established behavior for this family):**

- Keystroke logging
- Credential harvesting from browsers, email clients (Outlook), and FTP clients
- Clipboard monitoring
- Screenshot capture
- Exfiltration via SMTP email, FTP, Telegram Bot API, or HTTP-based C2 panels (varies by build)

## 4. Technical Behavior (Simulated Detonation Findings)

*Fields below should be completed from the sample's actual VirusTotal Behavior / Relations tabs. Structure reflects typical AgentTesla behavior for reference.*

| Category               | Finding                                             |
|------------------------|-----------------------------------------------------|
| Process injection      | <fill in — e.g. injected into RegAsm.exe / vbc.exe> |
| Persistence mechanism  | <fill in — e.g. Registry Run key, HKCU\...\Run>     |
| Exfiltration channel   | <fill in — SMTP server / Telegram API endpoint>     |
| Contacted C2 domain/IP | <fill in from Relations tab>                        |
| Dropped files          | <fill in from Behavior tab>                         |

## 5. Indicators of Compromise (IOCs)

| Type                 | Value                                                            |
|----------------------|------------------------------------------------------------------|
| SHA256               | 1c72369e2543722e89be3a14448d0fdc1664f4638d91d67720dddb7ff73cb736 |
| Family               | AgentTesla                                                       |
| C2 Domain/IP         | <fill in from Relations tab>                                     |
| Persistence artifact | <fill in from Behavior tab>                                      |

## 6. Impact to HRMC (Simulated)

**Confidentiality impact:** Stolen local credentials could enable follow-on lateral movement toward systems holding ePHI, directly relevant to the project's ePHI boundary rules.

**Risk register linkage:** Maps to an existing endpoint-compromise entry in the risk register (R-##) if one is already defined; if not yet captured, this incident supports adding a new risk entry for “credential theft via endpoint malware.”

**Control objective linkage:** Ties to the control objective(s) (CO-##) covering endpoint protection/EDR, and to the IAM control set — MFA reduces the value of stolen static credentials, tying back directly to the Keycloak MFA configuration.

## 7. Response Actions (Simulated / Planned)

1. Isolate affected endpoint from network (VLAN 30 quarantine).
2. Force credential reset for any accounts used on that workstation, prioritizing accounts with access to ePHI systems.
3. Review SIEM logs for the exfil domain/IP to check for other affected hosts.
4. Submit hash/IOC to threat intel feed for organizational blocklist update.

## 8. MITRE ATT&CK Technique Mapping (Preliminary)

| Technique                                              | ATT&CK; ID |
|--------------------------------------------------------|------------|
| Phishing                                               | T1566      |
| Credentials from Password Stores                       | T1555      |
| Input Capture: Keylogging                              | T1056.001  |
| Exfiltration Over Alternative Protocol (SMTP/Telegram) | T1048      |
| Scheduled Task/Job or Registry Run Keys (Persistence)  | T1547.001  |

This table feeds directly into Part 3 APT group research — cross-reference these techniques against candidate Groups' technique lists on the ATT&CK Enterprise matrix to identify which documented threat actor(s) align with the observed behavior chain.

---

*Prepared as part of the Care Access Navigator IAM & GRC Healthcare Cybersecurity Capstone. All hospital details, asset IDs, and incident data are simulated/synthetic for coursework purposes. Sample hash and detection ratio reference a real, publicly catalogued malware sample used solely as a technical grounding point.*